

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS2

SUBJECT: Biology

TOPIC: Microorganism around us

Definition of Microorganisms

Microorganisms or microbes are microscopic organisms, mostly unicellular or acellular that are too small to be seen with the naked human eye. They require a light or electron microscope for observation.

They are ubiquitous (found everywhere), in soil, aquatic environments (fresh water and marine), air, extreme environments (hot springs, deep ocean vents) and as endosymbionts or parasites inside other organisms. The branch of biology that entails the study of microorganisms is known as **MICROBIOLOGY**

Classification of Microorganisms

Microorganisms are grouped based on cellular structure (prokaryotic, eukaryotic, or non-cellular), nutritional modes, and genetic material.

1. Bacteria: They are microorganism that can be seen with a light microscope.
 - a. Cell Structure: Unicellular prokaryotes (lacking a membrane bound nucleus and organelle structures).
 - b. Cell Wall Composition: Peptidoglycan (murein).
 - c. Morphological Shapes: Cocci (spherical) e.g *Staphylococcus aureus*, *Bacilli* (rod shaped) *Bacillus subtilis*, *Escherichia coli*, *Spirilla* (spiral/twisted) e.g *Treponema pallidum*, *Vibrio* (comma shaped) e.g *Vibrio cholerae*.
2. Viruses: They exist on the biological borderline between the living and non-living world. They are acellular entities (they lack cellular structures like a cytoplasm, cell membrane or ribosomes).
They cannot perform metabolic processes or reproduce on their own, they are classified as obligate intracellular parasite: they must invade a living

host cell to replicate. Outside a host, they behave like non-living particles. Examples include Human Immunodeficiency Virus (HIV), Influenza Virus, Rabies Virus, Bacteriophages.

3. Microscopic Fungi

- i) Cell Structure: Eukaryotic (containing a true nucleus and bound organelles), can be unicellular (yeasts) or multicellular filament forming structures (molds)'
- ii) Cell Wall Composition: Chitin
- iii) Nutrition: Heterotrophic (saprophytic or parasitic), they secrete extracellular enzymes to digest organic substrates before absorption.
- iv) Examples include: *Saccharomyces cerevisiae* (Baker's yeast), *Rhizopus stolonifera* (Bread mold), *Penicillium chrysogenum*.

4. Protozoa

- i. Cell Structure: Unicellular, heterotrophic eukaryotes.
- ii. Locomotion: Grouped by motility mechanisms:
 - Pseudopodia* (false feet) e.g *Amoeba proteus*
 - Flagella* (whip-like structures) e.g *Trypanosoma gambiense*
 - Cilia* (hair-like structures) e.g *Paramecium caudatum*
 - Non-motile* e.g *Plasmodium falciparum*.

5. Microscopic Algae

- i. Cell Structure: Unicellular or simple colonial photosynthetic eukaryotes containing chlorophyll
- ii. Nutrition: Autotrophic (photosynthetic)

Meaning and Principles of Culturing Microorganisms

Microbial Culturing is the controlled laboratory technique of growing and multiplying microorganisms in designated nutrient media under favorable environmental conditions (temperature, pH, oxygen levels).

Concepts in Culturing

Culture Medium: A nutrient liquid broth or solid gel formulated with essential carbon, nitrogen, mineral salts, and vitamins required for microbial growth.

Agar: A complex polysaccharide extracted from red algae, used as a solidifying agent in culture media due to its inertness to microbial digestion.

Aseptic Technique: Sterilization protocols (e.g., autoclaving at 121°C/15 psi, flaming inoculation loops) used to prevent unwanted microbial contamination.

Pure Culture: A culture medium containing only a single, isolated species or strain of microorganism.

Importance & Applications of Culturing

- 1. Medical Diagnostics:** Isolating and identifying pathogenic bacteria or fungi from patient samples (blood, urine, swab) to determine antibiotic sensitivity.
- 2. Pharmaceutical Production:** Mass cultivation of Penicillium species for commercial antibiotic extraction.
- 3. Industrial Fermentation:** Industrial culturing of Saccharomyces for brewing and baking, or Lactobacillus for dairy products (yogurt, cheese).

Pathogens, Carriers, and Transmitting Organisms

Microorganisms that cause diseases in plants and animals are called **pathogens**. Pathogens rely on specific reservoirs and transmission vectors to spread between hosts.

Disease Carriers

A **carrier** is an infected individual (human or animal) who harbors a specific pathogen without showing overt clinical signs or symptoms, yet remains capable of transmitting the pathogen to susceptible individuals.

Asymptomatic Carrier: Harbors the pathogen continuously without experiencing active illness (e.g., individuals carrying Salmonella typhi causing Typhoid fever).

Convalescent Carrier: Continues to shed and transmit the pathogen while recovering from an infection.

Vectors & Transmitting Organisms

A **vector** is an intermediate host organism (usually an arthropod) that transfers a pathogenic microorganism from an infected host or reservoir to a susceptible host.

Vector Type	Transmission Mechanism	Vector Example	Pathogen Carried	Disease Caused
Biological	Pathogen multiplies/develops inside vector's	Female Anopheles mosquito	Plasmodium falciparum (Protozoan)	Malaria

	body; transmitted via bite.			
Biological	Pathogen transmitted directly into bloodstream via blood-feeding bite.	Tsetse fly (Glossina)	Trypanosoma species (Protozoan)	African Sleeping Sickness
Biological	Viral transmission via blood feeding bite.	Aedes aegypti mosquito	Flavivirus	Yellow Fever / Dengue
Mechanical	Pathogen attaches externally to legs/body from sewage/decay and contaminates food.	Housefly (Musca domestica)	Vibrio cholerae (Bacteria)	Cholera
Mechanical	External physical transfer of pathogens onto uncovered food and surfaces.	Cockroach (Periplaneta americana)	Entamoeba histolytica / Salmonella	Amoebic Dysentery / Food Poisoning

Assignment

1. Draw a simple structural diagram comparing the four main groups of microorganisms. Label at least one distinct feature for each group (e.g Capsid, cell wall material locomotion structure).
2. Differentiate clearly between a biological vector and a mechanical vector using their mode of pathogen transfer and one specific organism. Example for each.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

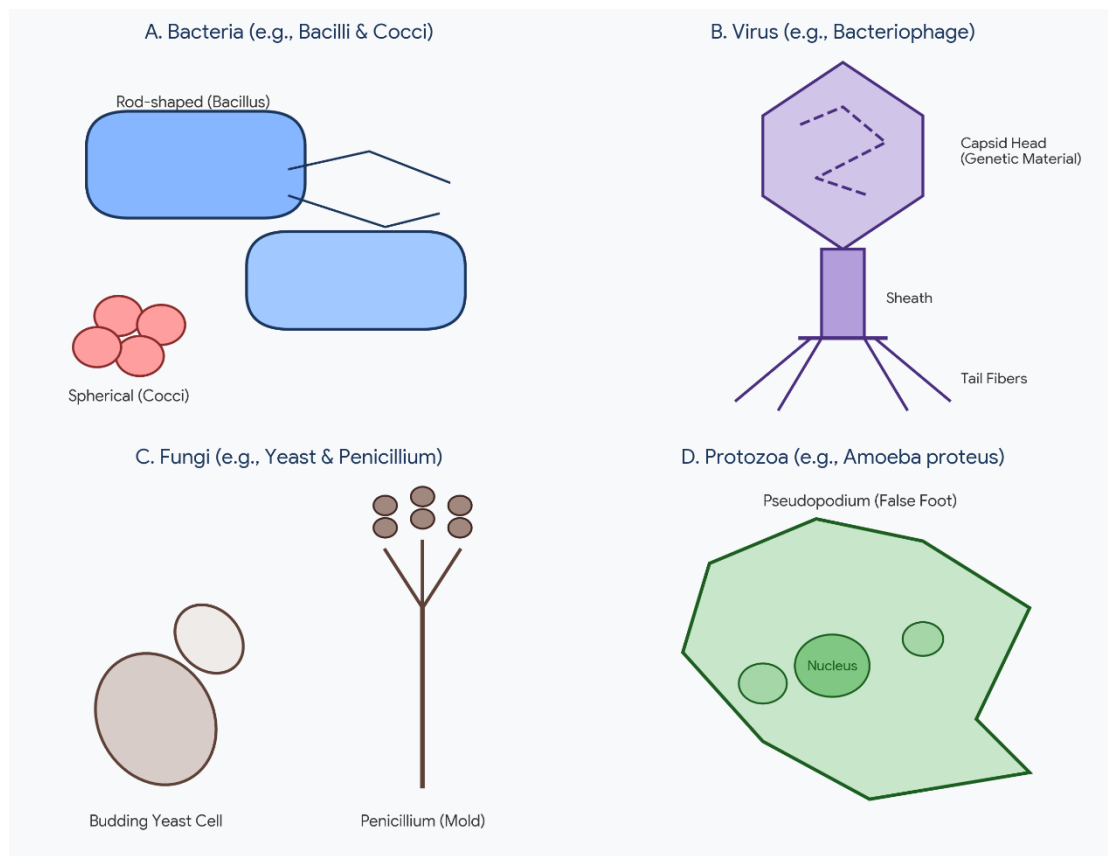
CLASS: SS2

SUBJECT: Biology

TOPIC: Microorganism in action and better health

Beneficial and Harmful Effects of Microorganisms

Microorganisms (microbes) are microscopic organisms such as bacteria, viruses, fungi, protozoa, and microscopic algae. While many are pathogenic (disease-causing), numerous microbes play essential roles in medicine, agriculture, food processing, and environmental sustainability.



Major groups of microorganisms and their characteristic structures.

Beneficial Effects of Microorganisms

- I. **Medicine & Health:**
 - Antibiotic Production: Fungi such as *Penicillium chrysogenum* yield penicillin; *Streptomyces* species produce streptomycin.
 - Vaccines & Sera: Attenuated (weakened) or killed microbes stimulate active immunity against severe pathogens.
 - Gut Microbiota: Normal intestinal bacteria (e.g., *Escherichia coli*) synthesize Vitamin K and Vitamin B complex while suppressing pathogenic invaders.
- II. **Agriculture & Soil Fertility:**
 - Nitrogen Fixation: Rhizobium bacteria in legume root nodules and free-living Azotobacter convert atmospheric nitrogen into soil nitrates.
 - Decomposition & Nutrient Cycling: Saprophytic bacteria and fungi break down dead organic matter, recycling essential elements (carbon, nitrogen, phosphorus).
- III. **Industry & Biotechnology:**

Food & Beverage Processing: *Saccharomyces cerevisiae* (yeast) ferments sugars into ethanol and CO₂ (baking and brewing). Lactobacillus species ferment lactose to lactic acid for yogurt and cheese production.

Sewage Treatment: Aerobic bacteria digest dissolved organic wastes in municipal sewage purification plants.

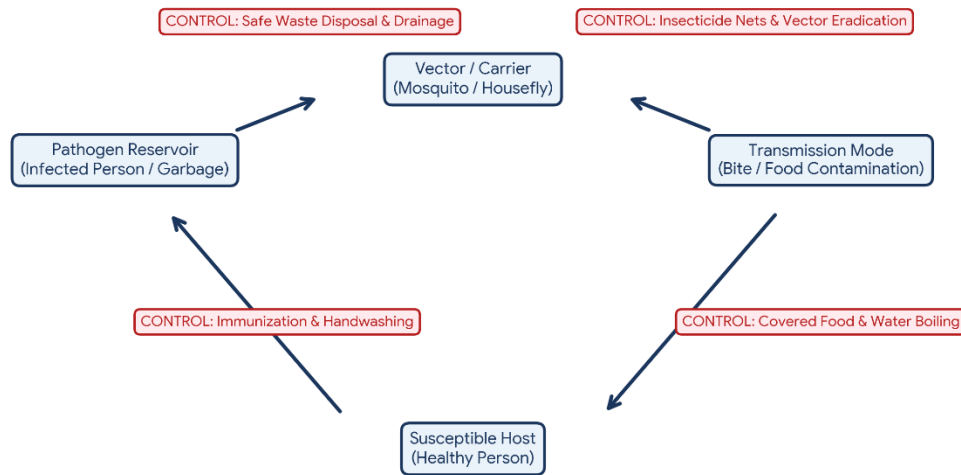
Harmful Effects of Microorganisms

- 1. Human & Animal Diseases:** Pathogens cause infections such as cholera (*Vibrio cholerae*), tuberculosis (*Mycobacterium tuberculosis*), malaria (*Plasmodium spp.*), and influenza.
- 2. Plant Diseases:** Microbes cause significant crop damage, e.g., cassava mosaic virus, maize rust, and potato late blight (*Phytophthora infestans*).
- 3. Food Spoilage:** Saprophytic bacteria and molds decompose raw and cooked food. *Clostridium botulinum* releases lethal neurotoxins causing botulism.
- 4. Material Deterioration:** Cellulolytic bacteria and fungi degrade timber, paper, textiles, and leather goods.

Control of Harmful Microorganisms & Public Health

Controlling microbial growth eliminates pathogens and halts disease transmission cycles within communities. Methods are categorized into physical, chemical, and public health measures.

Pathogen Transmission Cycle & Control Interventions



Disease transmission cycle and critical public health intervention points.

Physical Control Methods

High Temperature: Boiling water (100°C for 10–15 mins) kills vegetative bacterial cells. Autoclaving uses moist steam under pressure (121°C for 15 mins) to destroy heat-resistant bacterial endospores.

Pasteurization: Heating milk/liquids to 72°C for 15 seconds followed by rapid cooling eliminates pathogens without destroying flavor or nutritional value.

Low Temperature & Dehydration: Refrigeration (4°C) and freezing (-18°C) exert a bacteriostatic effect (halting growth). Desiccation removes moisture necessary for enzymatic activity.

Radiation: Ultraviolet (UV) radiation and gamma rays sterilize surgical instruments, pharmaceuticals, and packaged food.

Chemical Control Methods

Disinfectants: Strong chemical agents applied to non-living inanimate surfaces to destroy microbes (e.g., bleach, Lysol, phenol).

Antiseptics: Mild chemical solutions safe for application on living tissue and wounds to prevent infection (e.g., 70% ethanol, iodine, hydrogen peroxide).

Preservatives: Table salt, sugar, and vinegar inhibit microbial activity via osmotic dehydration or by lowering pH.

Public Health Measures & Vector Control

Safe Water Provision: Purification of municipal water through sedimentation, filtration, and chlorination.

Proper Waste Disposal: Underground sewage systems, sanitary pit latrines, and managed landfills prevent insect vector breeding.

Vector Eradication: Use of Long-Lasting Insecticide-treated Nets (LLINs) and indoor spraying against mosquitoes; food protection against houseflies (*Musca domestica*).

Immunization Programs: Routine mass vaccination builds herd immunity against polio, measles, diphtheria, and yellow fever.

Quarantine & Isolation: Enforced restriction of infected individuals or animal products during epidemic disease outbreaks.

Maintenance of Good Health

Good health is a state of complete physical, mental, and social well-being.

Maintaining health relies on three primary pillars:

Personal Hygiene: Frequent handwashing with soap under clean running water (especially before meals and after toilet use). Daily bathing, oral hygiene twice daily, and keeping fingernails trimmed short to eliminate dirt and parasite eggs.

Environmental Sanitation: Keeping living spaces, kitchens, and toilets clean and well-ventilated. Disposing of domestic refuse in covered bins and draining stagnant water pools to prevent mosquito breeding.

Healthy Lifestyle Factors: Consuming a balanced diet rich in essential micronutrients (Vitamin C, Zinc) to fortify immunity. Engaging in regular physical exercise and ensuring 7–8 hours of restful sleep daily for tissue recovery.

Roles of Health Organizations

1. World Health Organization (WHO): This is a specialized division of the United Nations Organization (UNO), now known as the United Nations, established in 1948 with headquarters in Geneva, Switzerland. Its major aim is to improve the health of the people in all the countries of the world.

2. United Nations Children's Fund (UNICEF): The UNICEF, another special agency of the United Nations Organization, now known as the United Nations, was set up to improve the health and welfare of the children all over the world.

3. Nigerian Medical Association (NMA)

The Nigerian Medical Association (NMA) is a national body concerned with the maintenance of good health within Nigeria.

Assignment

1. Outline three public health interventions required to prevent an outbreak of cholera in a flood-affected community.

2. Describe how pasteurization differs from standard boiling in terms of temperature, duration, and effect on liquid nutrients.
3. List three personal hygiene practices and explain how each prevents pathogen invasion into the human body.

Evaluation(student)

1. Identify the beneficial and harmful effects of some microbes.
2. Describe physical and chemical methods used to control harmful microorganisms.
3. Outline public health measures and control strategies implemented to prevent infectious microbial diseases.
4. Enumerate essential personal hygiene practice for maintaining good health.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS2

SUBJECT: Biology

TOPIC: Aquatic Habitat (Marine Habitats)

Procedure (content development)

Characteristics/features of Marine Habitat

A **marine habitat** is an aquatic environment composed of saltwater, encompassing all oceans and seas. It covers approximately **70%** of the earth's surface and serves as the largest global aquatic biome, supporting diverse ecosystems.

Characteristics of Marine Habitat

The marine or salt water habitats possess the following key physical and chemical characteristics:

1. Salinity: Salinity is defined as the degree of saltiness or concentration of salt solution in oceans. Marine habitats have high salinity, with an average salinity put at 35.2 per 1000 (35.2 parts of salts by weight per 1000 parts of water).

2. Density: The density of marine water is high (~1.028) compared to fresh water (1.00). Hence, many organisms can easily float on ocean water.

3. Pressure: Water pressure increases with depth at the rate of one atmosphere for every 10 meters, varying from 1 atm at the surface to about 1000 atm at extreme depths.

4. Size: Marine habitats represent the largest of all habitats. Oceans alone occupy over 70% (approx. 360 million km²) of the Earth's total surface area (510 million km²). Examples include the Atlantic, Indian, and Pacific Oceans.

5. Currents: Currents are produced by wind action at the surface and by internal variations in salinity and temperature deeper in the ocean.

6. Tides: Tides are the periodic rise and fall of the ocean surface (occurring roughly twice daily), caused by the gravitational attraction of the moon and sun.

7. Oxygen Concentration: Highest at the surface level where atmospheric exchange occurs; oxygen concentration decreases with depth, becoming practically non-existent in extreme depths.

8. Hydrogen Ion Concentration (pH): Seawater is alkaline in nature with a typical pH ranging between 8.0 and 9.0 near the surface.

9. Waves: Movements of surface waters caused by winds that facilitate continuous mixing of seawater near the surface.

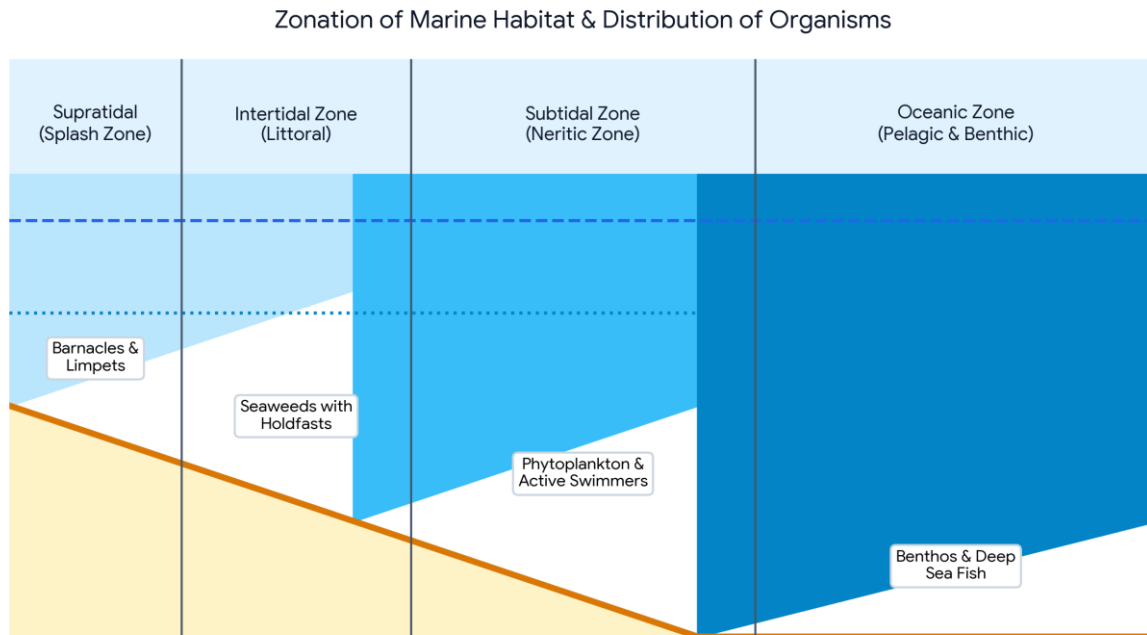
10. Light Penetration: Light penetrates ocean water up to a maximum depth of about 200 meters (Euphotic zone), directly limiting plant life to these upper illuminated layers.

Major Zones of the Marine Habitat

Supratidal (Splash) Zone	Area above the high-tide mark receiving moisture from ocean splash and wave action.
Intertidal (Neritic) Zone	Area between high and low tides, characterized by high light penetration, fluctuating water temperatures, and continuous wave movement.
Littoral (Subtidal) Zone	Extends down to roughly 200m depth; continuously submerged, receives abundant sunlight, and contains rich nutrient supplies.
Benthic Zone	Extends down to about 500m depth; characterized by low light penetration and low nutrient availability.
Pelagic / Abyssal Zone	Reaches depths of about 7000m; cold, high pressure, low light penetration, and low primary organic productivity.

Hadal (Aphotic) Zone

Deepest marine zones (>7000m depth) forming ocean trench beds with absolute darkness and no photosynthetic activity.



Distribution of organisms in habitats

Organisms in marine environments are distributed into three main ecological groups:

1. **Plankton (Floating organisms):** Drifting microscopic organisms found mainly in the lighted upper zone.
 - Phytoplankton (Producers):** Microscopic plant-like algae (e.g., Diatoms, Dinoflagellates) that carry out photosynthesis.
 - Zooplankton (Consumers):** Microscopic animals (e.g., Protozoa, Copepods, jellyfish larvae).
2. **Nekton (Active swimmers):** Animals capable of swimming strongly against water currents (e.g., Tilapia, Tuna, Sharks, Whales, Sea Turtles, Squids).
3. **Benthos (Bottom dwellers):** Organisms attached to or crawling on the sea floor (e.g., Sea Anemones, Corals, Sea Urchins, Crabs, Starfish, Sea Cucumbers).

Adaptive features of marine organisms

Plant Adaptations

Salt Water Swamp Plants: Possess succulent stems/roots to dilute salt concentration.

Have pneumatophores (breathing roots) for atmospheric gas exchange.

Stilt roots provide structural anchorage in mud.

Viviparous seedlings ensure survival and continuity.

Excrete excess salt to maintain internal body salt balance.

Seaweeds: Holdfast structures for firm attachment to rocks.

Mucilaginous outer layer prevents desiccation.

Divided leaves and air bladders for buoyancy in water.

Animal Adaptations

Barnacles: Possess protective mantles or attachment pads to firmly anchor onto rocks, preventing desiccation during low tide.

Cartilaginous Fishes (e.g., Sharks, Dogfish): Maintain high internal salt balance and body osmotic pressure to adapt to high ocean salinity.

Bony Fishes (e.g., Tilapia, Herring): Drink salt water to cope with high salinity and utilize salt-secreting glands in their gills or eyes to excrete excess salt.

Shrimps: Equipped with powerful chelipeds (claws) for capturing and holding prey.

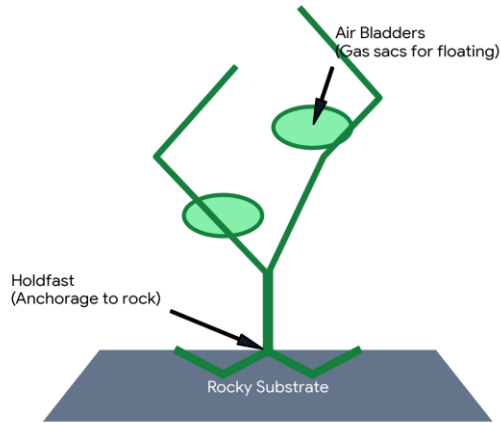
Crabs: Burrow rapidly into coastal mud to protect against wave action, tides, and predators.

Periwinkles: Possess modified lungs for breathing air during exposure and a strong muscular foot for firm attachment.

Starfish: Utilize tube feet to hold securely onto rocky shores and hard outer shells to prevent desiccation.

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Marine Plant Adaptations (Seaweed / Sargassum)



Marine Animal Adaptations (Bony Fish)

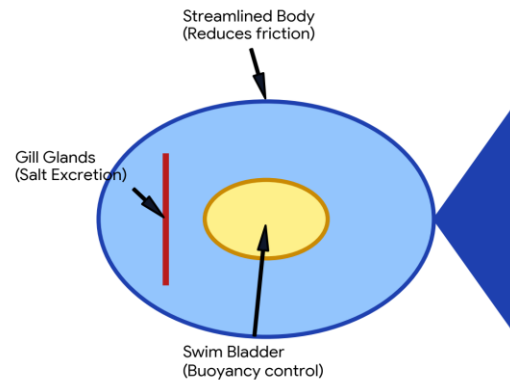


Diagram showing Adaptive Features of Marine Plants (Holdfast & Air Bladder) and Animals (Swim Bladder & Gills)

Assignment

1. Mention five threats to marine habitats.
2. Explain five ways of conserving marine habitat.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS2

SUBJECT: Biology

TOPIC: Estuarine Habitats

Types of Estuarine Habitat

An **estuarine habitat** (or estuary) is a semi-enclosed coastal body of water where fresh water from rivers and streams meets and mixes with salt water from the ocean. This mixing produces **brackish water**, which has a salinity intermediate between fresh water (<0.5‰) and full seawater (~35‰).

Estuaries are highly dynamic transition zones (ecotones) affected by tidal fluctuations, high nutrient loads, and fluctuating environmental parameters.

Types of Estuaries

- 1. Delta:** A delta is where a river divides into many channels before entry into the ocean or sea. Brackish water or estuary (delta) is formed at the mouth of a river as it enters the sea.
- 2. Lagoon:** This is a body of ocean water that enters into the land through a canal and therefore has the opportunity of mixing with fresh water from rivers and streams.
- 3. Bay:** This is a little or small portion of the sea water which enters into the land and mixes up with fresh water from rivers and streams. A bay is very small and not long enough to rejoin the sea at another end.

Characteristics of Estuarine Habitats

The following are the principal characteristics of estuarine habitats:

- (1) Fluctuation in salinity:** Salinity fluctuates in this habitat. Salinity is lower at the mouth of a river and gets higher towards the sea. Salinity is also affected by season: while the rainy season reduces salinity due to the addition of fresh water, the dry season increases it.

(2) Turbidity: Turbidity of estuarine habitats increases especially during the rainy season when lots of debris are brought down by rivers to the habitat. This high turbidity also reduces the rate of photosynthesis and respiration by organisms.

(3) Shallowness of water: Unlike sea water which is deep, the water in an estuarine habitat is generally very shallow.

(4) Low species diversity: The estuarine habitat has low diversity of species compared to marine habitats. Common plant species include phytoplankton, algae, and marsh vegetation, while animal species include crabs, oysters, lobsters, etc.

(5) Water is affected by tides: Sea water usually flows rapidly into estuaries at high tides and rushes back into the ocean at low tides.

(6) High level of nutrients: The estuarine habitats contain abundant nutrients, especially organic detritus, which form the bulk of producers in the habitat.

(7) Low oxygen content: Oxygen content of estuarine habitats is generally very low and, as a result, much of the microbiological activity is anaerobic.

Distribution of Plants and Animals in Estuarine Habitats

Plant Species and Their Adaptive Features

(1) Planktons (diatoms): They possess air spaces in their tissues, rhizoids or false feet for attachment to rocky shores, and air bladders for buoyancy.

(2) Algae: They possess chlorophyll for photosynthetic activities and a small size or large surface-area-to-volume ratio for floating.

(3) Red mangrove (*Rhizophora racemosa*): It has stilt roots which grow down from the stem into the soft mud and develop numerous rootlets with air spaces for conducting air to root tissues. The roots also provide support and prevent plants from being washed away by tides. Furthermore, the seeds of red mangrove germinate while still on the parent plant (vivipary), preventing seedlings from being carried away by water currents.

(4) White mangrove (*Avicennia nitida*): It has pneumatophores (breathing roots) for gaseous exchange.

Animal Species and Their Adaptive Features

(1) Mosquito larvae and pupae: These organisms possess breathing trumpets for gaseous exchange.

- (2) Crabs:** They can burrow fast into the mud for protection against predators, strong waves, or tides.
- (3) Water snails and shrimps:** These animals can burrow into the mud when the tide is going out, thus escaping the periodic dilution of their external medium.
- (4) Worms:** They have strong, protective, impermeable coverings against high salinity.
- (5) Mudskippers:** These animals have modified fins adapted for crawling when on land and for swimming when in water.
- (6) Fishes:** Fishes like Tilapia have fins for movement and a swim bladder for buoyancy control.

Food Chain in Estuarine Habitats

A typical food chain in an estuarine habitat may contain up to three, four, or five trophic levels. Phytoplankton (such as diatoms) and organic detritus form the primary producers supporting the food chain.

Examples of Food Chains in Estuarine Habitats:

- (i) Detritus → Worms → Snails → Birds
- (ii) Diatoms → Shrimps → Fishes
- (iii) Phytoplankton → Barnacles → Fish → Birds

Factors Affecting Estuarine Habitats

The environmental factors affecting estuarine habitats are common to most aquatic ecosystems and include:

Temperature

Relative humidity

Light intensity & penetration

pH (Acidity / Alkalinity).

Assignment

1. Describe the estuarine habitat under the following headings:

- a) Meaning of estuarine habitat
- b) Characteristics
- c) Plants and animals found there
- d) Adaptations of organisms
- e) Economic importance

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RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS2

SUBJECT: Biology

TOPIC: Fresh water Habitats

Types and characteristics of fresh water habitat

Fresh water habitat is a body of water formed mainly from inland waters and contain a very low level of salinity. Examples of fresh water habitat are rivers, ponds, streams, springs and lakes. Freshwater habitats are classified primarily based on the mobility or movement direction of the water body into two distinct categories: **Lotic (Running) Waters** and **Lentic (Stagnant) Waters**.

Lotic (Running) Waters: Waters that flow continuously in a specific direction. They possess strong water currents, high oxygen concentration due to constant aeration, and are generally non-stratified. Examples are Rivers, streams, springs, brooks, waterfalls.

Lentic (Stagnant) Waters: Standing or calm waters that do not flow in a specific direction. They accumulate organic sediments, undergo thermal/light stratification, and exhibit distinct zonation. Examples are Lakes, ponds, swamps, dams, reservoirs, bogs.

Characteristics of Fresh Water Habitats

Low Salinity: Freshwater habitats contain very low levels of dissolved salts (typically around 0.5% or less), compared to oceans which average about 3.5% (35 parts per thousand).

Small Relative Size: Freshwater habitats occupy a very small proportion of Earth's total water surface compared to marine habitats, which cover over 75% of the Earth's surface.

Temperature Variations: Temperature fluctuates with seasonal changes, weather, and depth. In lentic bodies, surface water is slightly warmer than bottom mud during warm periods.

High Dissolved Oxygen Content: Oxygen is readily available in most freshwater zones, particularly in fast-flowing lotic waters and the surface layer of lentic bodies.

Shallowness: Most freshwater habitats are relatively shallow compared to marine oceans, allowing light to easily penetrate to the substrate in many areas.

Seasonal Volume Variation: Streams and rivers experience high volume, swift currents, and increased turbidity during rainy seasons, but dry up or shrink significantly in dry seasons.

Current Action: Unidirectional currents in lotic environments distribute dissolved gases, salts, nutrients, and microorganisms while acting as a physical challenge to non-adapted organisms.

Ecological Zones of Fresh Water Habitats

1. Lentic Habitat Zones (e.g., Lakes and Ponds)

Lentic habitats display two primary vertical ecological zones based on light penetration, rooted vegetation, and depth:

Littoral Zone: The shallow edge/margin of the water body where sunlight easily penetrates to the bottom sediment. It features rooted vegetation, warm temperatures, maximum dissolved oxygen, and the highest primary biological productivity.

Plants: Spirogyra, Chlamydomonas, water fern, duckweed, water lettuce (Pistia), diatoms, sedges.

Animals: Water fleas, water snails, flatworms, frogs, toads, water skaters (Gerri's), ducks, snakes, crocodiles, tadpoles, Hydra, hippopotamus.

Benthic Zone: The deep bottom region composed of organic mud and substrate (humus). Although rooted vegetation like littoral plants is absent at the base, floating plants with deep-anchored roots exist here. Light is limited, and organic decomposition is high.

Plants: Water lily (Nymphaea), water arum, ferns, Crinum lily, Commelina, submerged water grasses.

Animals: Protozoa, rotifers, Hydra, tilapia fish, mudfish, catfish, leeches, caddisfly larvae, mosquito larvae/pupae, water spider, crayfish, water scorpion, water boatman, water bugs.

2. Lotic Habitat Zones (e.g., Rivers and Streams)

In lotic (running) water ecosystems, zones are differentiated by water flow velocity and pool depth:

Pool Zone: Deeper regions of the river where water flow is relatively slow, calm, and peaceful. Fine silt and organic matter settle here.

Rapid Zone: Shallow areas where water flows swiftly over rocky substrates. Aeration is maximum, but thermal stratification does not occur due to continuous mixing.

Eutrophication in Freshwater Ecosystems

Eutrophication is defined as the excessive growth of plant life or algal blooms in a body of water, caused by an enrichment of organic nutrients and minerals, ultimately leading to severe oxygen depletion and the collapse of aquatic life.

Major Causes of Eutrophication:

Overuse of agricultural fertilizers containing nitrates and phosphates, which wash into rivers/lakes during heavy rainfall.

Discharge of untreated domestic sewage and organic waste runoff from fish farming operations.

Adaptive Features of Freshwater Organisms

Organisms inhabiting fresh waters face challenges such as osmotic water gain (due to hypotonic water), strong water currents in lotic habitats, and light limitations in deep lentic habitats. Below are their key structural, physiological, and behavioral adaptations.

Adaptive Features of Freshwater Plants

Plant Organism	Structural / Adaptive Feature	Ecological Survival Function
Water lily (Nymphaea)	Air bladders, broad expanded floating leaves, light weight, long central petioles attached to base mud.	Air bladders and broad leaves keep the plant afloat. Long petioles prevent leaves from being pulled underwater by water currents.

Plant Organism	Structural / Adaptive Feature	Ecological Survival Function
Water hyacinth (Ipo)	Spongy leaf stalks with numerous cavities and large intercellular air spaces.	Traps air within tissues to provide buoyancy and float on the water surface.
Spirogyra	Outer protective mucilaginous sheath covering the filamentous body.	Protects the algae from decay, reduces friction, and prevents physical damage from water currents.
Water lettuce (Pistia)	Dense, fine microscopic hairs covering the leaf surface.	Traps air bubbles between hairs, making the leaves water-repellent and helping the plant float.
Hornwort (Ceratophyllum)	Submerged, highly dissected, thin thread-like leaves.	Increases surface area for maximum sunlight absorption and efficient gaseous exchange underwater.
Water weed (Elodea)	Long, slender, flexible submerged petioles and stems.	Enables the plant to bend gracefully and flex with water currents without breaking.

Adaptive Features of Freshwater Animals

Animal Organism	Structural / Adaptive Feature	Ecological Survival Function
Protozoa (Amoeba/Paramecium)	Contractile vacuoles.	Performs osmoregulation by collecting and expelling excess water entering the cytoplasm via osmosis.
Tilapia Fish	Hydrodynamic body shape, internal swim bladder, highly vascularized gills.	Swim bladder controls buoyancy; gills enable underwater gaseous exchange; streamlined shape minimizes friction.
Duck	Webbed feet digits; serrated filtering beak.	Webbed digits act as paddles for swimming; serrated beak sieves plankton and small food particles from mud.
Lungfish (Protopterus)	Dual breathing organs (gills & functional lungs); ability to secrete mucus cocoon.	Uses gills in water; uses lungs to breathe atmospheric air during dry seasons by burrowing into mud (estivation).
Hydra	Slippery body outer layer, adhesive foot disc, hooks, suckers, and nematocysts.	Suckers and sticky disc enable secure attachment to water plants; nematocysts capture prey.
Pond skaters (Gerris)	Long slender legs equipped with water-repellent hair fringes.	Distributes weight to skate across water surface tension without breaking through.

Animal Organism	Structural / Adaptive Feature	Ecological Survival Function
Water boatman	Hydrofuge body hairs that trap air bubbles beneath wings.	Carries a bubble of atmospheric air underwater to serve as a physical gill for respiration.

Comparative Adaptations: Stagnant (Lentic) vs. Running (Lotic) Waters

Ecological Adaptation Category	Stagnant Water Adaptations (Lentic Ecosystems)	Running Water Adaptations (Lotic Ecosystems)
Plant Adaptations	• Broad floating leaves with air bladders (Water lily, Hyacinth). • Fine leaf hairs to trap air (Water lettuce). • Flexible stems anchored in soft mud.	• Flexible, stream-line dissected leaves (Hornwort, Elodea) to resist tearing. • Strong root anchoring systems in gravel/rocks. • Thick mucilage coating (Spirogyra).
Animal Adaptations (Morphology)	• Surface tension skimmers (Pond skaters). • Air-holding structures (Water boatman air bubble). • Buoyancy organs (Swim bladders).	• Flattened bodies to slip under rocks. • Suckers, hooks, sticky foot discs (Hydra, Leeches). • Strong muscle swimmers with streamlined bodies.
Respiration & Survival	• Gills & Lungs for estivation in mud (Lungfish). • Contractile vacuoles for osmoregulation.	• Highly developed gill surface area to absorb abundant dissolved oxygen in rapids.

Food Chains and Energy Flow in Fresh Waters

Freshwater Food Chains

Diatoms (Producer) → Fish fry (Primary Consumer) → Tilapia (Secondary Consumer)

Detritus (Organic waste) → Worm (Primary Consumer) → Shrimps (Secondary Consumer) → Water Bird (Tertiary Consumer)

Spirogyra (Producer) → Tadpoles (Primary Consumer) → Carps (Secondary Consumer) → Kingfisher (Tertiary Consumer)

Energy Flow Pathway

Radiant Solar Energy is absorbed by green plants and phytoplankton (producers) for photosynthesis to synthesize organic carbohydrates.

Herbivores (zooplankton, tadpoles) consume producers directly (primary consumer level).

Carnivores (small/large fish, water birds) obtain energy indirectly by feeding on herbivores.

Decomposers (saprophytic fungi and bacteria) break down organic wastes and dead bodies, releasing inorganic nutrients back into the mud.

Energy Loss: At each trophic transfer level, a significant proportion of energy is lost as heat through respiration and metabolic activities.

Factors Affecting Fresh Water Habitats

Freshwater ecosystems are regulated by both biotic and abiotic ecological factors:

Biotic Factors: Producers (green plants, diatoms, algae), Consumers (fish, tadpoles, insects, aquatic birds), Parasites, and Decomposers (bacteria, saprophytic fungi).

Abiotic Factors: Water current velocity, temperature, dissolved oxygen content, salinity level, water turbidity/clarity, pH, and sunlight penetration depth.

Assignment

1. With the aid of a detailed flowchart/diagram, describe the step-by-step process of eutrophication and its ultimate impact on aquatic life.
2. Construct a food chain consisting of four trophic levels in a freshwater habitat and explain how energy is lost from producers to decomposers

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS2

SUBJECT: Biology

TOPIC: Terrestrial Habitats

SUB TOPIC: Features, Formation, types of Marshes and adaptation of

Features of Marsh

Terrestrial habitats refer to life on earth. The nature of soil, rainfall and temperature are the major factors affecting the nature of terrestrial habitats. Terrestrial habitat is divided into four groups namely: Marsh, Rainforest, Savanna or Grassland and Arid.

Marshy Habitats

Marshy habitat is defined as a lowland habitat which is usually flooded or water logged all the time. Naturally, grasses and shrubs grow in marsh but when trees grow there, it is called a **swamp**. Marsh is often regarded as a transition between the aquatic and terrestrial habitats.

Features of Marsh Habitats:

1. **Waterlogged & Anaerobic Soil:** The ground remains saturated with water for extended periods, creating an oxygen-depleted (anaerobic) underground environment that slows organic decomposition.
2. **Herbaceous Plant Dominance:** Trees are largely absent; non-woody plants with soft, flexible stems dominate the landscape.
3. **Shallow Water Level:** Water depth typically ranges from a few centimeters to about a meter, allowing sunlight to reach the substrate and support dense vegetation.
4. **High Organic Accumulation:** Partially decayed plant material accumulates to form nutrient-rich mud, silt, and peat deposits.
5. **Dynamic Salinity Levels:** Depending on geographical location, salinity can range from completely fresh to brackish or hypersaline coastal water

Formation of Marshes (Ecological Succession)

Stage 1: Sediment Accumulation: Slow moving rivers, streams, or coastal tides deposit fine silt, clay, and organic debris in low lying shallow water basins.

Stage 2: Shallowing of water: Continuous deposition reduces water depth, enabling sunlight to penetrate the basin floor and creating ideal conditions for plant attachment.

Stage 3: Pioneer Colonization: water tolerant herbaceous species (e.g. reeds, sedges, cattails) take root in the soft mud, Marshes develop gradually through a process of ecological succession in areas with poor drainage and continuous sediment input: stabilizing the sediment with their root networks.

Stage 4: Organic Maturation: As plant material dies and decays, organic peat accumulates, raising the ground level and establishing a mature marsh ecosystem.

Types of Marshes

There are two types of marshes. These are freshwater marshes and salt water marshes.

Freshwater marshes: Fresh water marshes occur on land, just beyond the limits of the salt water marshes and beyond the areas influenced by tide. In this area, fresh water from rivers overflows the river banks to flood the adjoining lowlands resulting in the formation of fresh water marshes.

Saltwater marshes: Saltwater marshes occur along the coastal areas and they are influenced by tides because the water along the coast is salty, it mixes up with fresh water from rivers to form brackish water. The action of tide in the ocean causes the flooding of adjoining lowlands with brackish water resulting in the formation of salt water marshes.

Typical Marsh Organisms and Adaptive Features

Organisms living in marsh habitats face severe environmental challenges, including waterlogging, oxygen-depleted mud, fluctuating water levels, and variable salinity. Below are key adaptive features of representative marsh flora and fauna:

Category	Organism	Structural / Physiological Adaptation	Survival Function
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Flora (Plants)	Cattail (Typha)	Spongy Aerenchyma Tissues	Transports atmospheric oxygen down to roots submerged in anaerobic mud.
Flora (Plants)	Water Lily (Nymphaea)	Swollen Air-filled Petioles	Provides buoyancy to floating leaves and prevents submergence.
Flora (Plants)	Cordgrass (Spartina)	Specialized Salt Glands	Excretes excess salt absorbed from brackish/salt water.
Fauna (Animals)	Frogs & Toads	Moist Skin & Webbed Feet	Enables cutaneous respiration underwater and efficient paddling in mud.
Fauna (Animals)	Mudskipper	Enlarged Water-Retaining Gill Chambers	Allows respiration out of water while crawling over mud banks.
Fauna (Animals)	Heron / Egret	Long Stilt Legs & Spread Toes	Wades through water without wetting plumage; toes prevent sinking in mud.

Assignment

Draw a large, neatly labeled diagram of a wading bird (e.g., Heron or Egret). Annotate at least THREE structural adaptations that enable it to thrive in a marsh habitat.

